

Second Quantization

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References:

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Permutation Symmetry

Second quantization is a convenient way to deal with identical particles. From introductory quantum mechanics, we understand it is not meaningful to label identical particles. However, to build a rigorous mathematical foundation for identical particles, we assume such labelling is possible. In other words, we first assume that every particle is different.

For N particles, the quantum state can be generally expressed as

$$|\Psi\rangle = \sum_{\alpha, \beta, \gamma, \dots} \Psi(\alpha, \beta, \gamma, \dots) |\alpha\rangle_1 |\beta\rangle_2 |\gamma\rangle_3 |\dots\rangle. \quad (1)$$

Here $|\alpha\rangle_1 |\beta\rangle_2 |\gamma\rangle_3 |\dots\rangle$ means particle 1 is in the state labeled by α , particle 2 is in the state labeled by β , etc. This is a direct product state. $|\alpha\rangle_1 |\beta\rangle_2 |\gamma\rangle_3 |\dots\rangle$ is a product state, and all possible product states span the Hilbert space of N particles.

The **permutation operator** $\hat{P}$ permutes the particles cyclicly. For example

$$\hat{P}_{12} |\alpha\rangle_1 |\beta\rangle_2 |\gamma\rangle_3 = |\alpha\rangle_2 |\beta\rangle_1 |\gamma\rangle_3 = |\beta\rangle_1 |\alpha\rangle_2 |\gamma\rangle_3, \quad (2)$$

and

$$\hat{P}_{123} |\alpha\rangle_1 |\beta\rangle_2 |\gamma\rangle_3 = |\alpha\rangle_2 |\beta\rangle_3 |\gamma\rangle_1 = |\gamma\rangle_1 |\alpha\rangle_2 |\beta\rangle_3. \quad (3)$$

For convenience, we can omit the subscript (the particle label) and assume particles are ordered. For example, $|\gamma\rangle |\alpha\rangle |\beta\rangle$ means particle 1 at state γ , particle 2 at state α , etc.

All permutations can be expressed as product of cyclic permutations. The permutation group S_N which consists of all permutations of N objects has $N!$ elements.

A few properties of the permutation operators are noted:

1. $\hat{P}_{ij}^2 = 1$, such that its eigenvalues are ± 1 .
2. $\hat{P}$ is a unitary symmetry: $\langle \hat{P}\Psi_i | \hat{P}\Psi_j \rangle = \langle \Psi_i | \Psi_j \rangle$.
3. For every symmetric operator $\hat{O}$, $\hat{P}\hat{O} = \hat{O}\hat{P}$. This can be viewed as the definition of an symmetric operator. In addition,

$$\langle \hat{P}\Psi_i | \hat{O} | \hat{P}\Psi_j \rangle = \langle \Psi_i | \hat{O} | \Psi_j \rangle. \quad (4)$$

All physical observable should correspond to symmetric operators, including the Hamiltonian.

4. If $|\Psi\rangle$ is an eigenstate of $\hat{H}$, $\hat{P}|\Psi\rangle$ is also an eigenstate of $\hat{H}$ with the same eigenvalue.

If we start with a product state $|\alpha\rangle|\beta\rangle|\gamma\rangle$ of three particles and act all possible permutation operators on it, we get

$$|\alpha\rangle|\beta\rangle|\gamma\rangle, |\alpha\rangle|\gamma\rangle|\beta\rangle, |\beta\rangle|\alpha\rangle|\gamma\rangle, |\beta\rangle|\gamma\rangle|\alpha\rangle, |\gamma\rangle|\alpha\rangle|\beta\rangle, |\gamma\rangle|\beta\rangle|\alpha\rangle. \quad (5)$$

It is possible to form invariant linear spaces of such states, such that all permutation in S_3 cannot mix such linear spaces. In group theory language, we are looking for irreducible representations. For example, for the above 6 states, the invariant linear spaces are:

1. Symmetric space:

$$\frac{1}{\sqrt{6}}(|\alpha\rangle|\beta\rangle|\gamma\rangle + |\alpha\rangle|\gamma\rangle|\beta\rangle + |\beta\rangle|\alpha\rangle|\gamma\rangle + |\beta\rangle|\gamma\rangle|\alpha\rangle + |\gamma\rangle|\alpha\rangle|\beta\rangle + |\gamma\rangle|\beta\rangle|\alpha\rangle) \quad (6)$$

2. Antisymmetric space:

$$\frac{1}{\sqrt{6}}(|\alpha\rangle|\beta\rangle|\gamma\rangle - |\alpha\rangle|\gamma\rangle|\beta\rangle - |\beta\rangle|\alpha\rangle|\gamma\rangle + |\beta\rangle|\gamma\rangle|\alpha\rangle + |\gamma\rangle|\alpha\rangle|\beta\rangle - |\gamma\rangle|\beta\rangle|\alpha\rangle) \quad (7)$$

3. Two-dimensional space 1:

$$\frac{1}{\sqrt{12}}(2|\alpha\rangle|\beta\rangle|\gamma\rangle - |\alpha\rangle|\gamma\rangle|\beta\rangle + 2|\beta\rangle|\alpha\rangle|\gamma\rangle - |\beta\rangle|\gamma\rangle|\alpha\rangle - |\gamma\rangle|\alpha\rangle|\beta\rangle - |\gamma\rangle|\beta\rangle|\alpha\rangle), \quad (8)$$

and

$$\frac{1}{2}(-|\alpha\rangle|\gamma\rangle|\beta\rangle - |\beta\rangle|\gamma\rangle|\alpha\rangle + |\gamma\rangle|\alpha\rangle|\beta\rangle + |\gamma\rangle|\beta\rangle|\alpha\rangle) \quad (9)$$

4. Two-dimensional space 2:

$$\frac{1}{\sqrt{12}}(2|\alpha\rangle|\beta\rangle|\gamma\rangle + |\alpha\rangle|\gamma\rangle|\beta\rangle - 2|\beta\rangle|\alpha\rangle|\gamma\rangle - |\beta\rangle|\gamma\rangle|\alpha\rangle - |\gamma\rangle|\alpha\rangle|\beta\rangle + |\gamma\rangle|\beta\rangle|\alpha\rangle), \quad (10)$$

and

$$\frac{1}{2}(-|\alpha\rangle|\gamma\rangle|\beta\rangle + |\beta\rangle|\gamma\rangle|\alpha\rangle - |\gamma\rangle|\alpha\rangle|\beta\rangle + |\gamma\rangle|\beta\rangle|\alpha\rangle). \quad (11)$$

The symmetric basis is the eigenstate of all $\hat{P}_{ij}$ with eigenvalue 1, and the antisymmetric basis is the eigenstate of all $\hat{P}_{ij}$ with eigenvalue -1 . It can be easily verified that the two two-dimensional spaces are closed under the operation of all possible permutations.

We now consider identical particles. *For identical particles, it is established that the relevant Hilbert space is either the symmetric space or the antisymmetric space.* The general way to construct symmetrized and antisymmetrized states is

$$\hat{S}_{\pm}|i_1, i_2, \dots, i_N\rangle = \frac{1}{\sqrt{N!}} \sum_P (\pm 1)^P \hat{P}|i_1, i_2, \dots, i_N\rangle, \quad (12)$$

where $|i_1, i_2, \dots, i_N\rangle$ is an abbreviation for $|i_1\rangle|i_2\rangle\dots|i_N\rangle$. We have abused the notation to use P to also denote the number of exchange operations ($i \leftrightarrow j$) needed to construct the permutation. $\hat{S}_+$ corresponds to symmetric state and $\hat{S}_-$ corresponds to antisymmetric state. Second quantization is a method to work with such states in a convenient way.

Some basic properties of $\hat{S}_\pm|i_1, i_2, \dots, i_N\rangle$:

1. If any two (or more) of the states in $\{i_1, i_2, \dots, i_N\}$ are the same, $\hat{S}_-|i_1, i_2, \dots, i_N\rangle=0$. This is **Pauli exclusion principle** for fermions. This can be proved by $\hat{P}\hat{S}_- = (-1)^P\hat{S}_-$.
2. $\hat{S}_+|i_1, i_2, \dots, i_N\rangle$ is not necessarily normalized. Let's assume there are n_1 repeated state 1, n_2 repeated state 2, etc,

$$\langle i_1, i_2, \dots, i_N | \hat{S}_+^\dagger \hat{S}_+ | i_1, i_2, \dots, i_N \rangle = n_1! n_2! \dots n_N!. \quad (13)$$

Fock space, Creation and Annihilation Operator

We first consider bosons, for which the $\hat{S}_+$ is relevant. The occupation number representation of a state is defined as:

$$|n_1, n_2, \dots\rangle = \hat{S}_+ |i_1, i_2, \dots, i_N\rangle \frac{1}{\sqrt{n_1! n_2! \dots}}. \quad (14)$$

Here, $|n_1, n_2, \dots\rangle$ means the first state is occupied by n_1 electrons, the second state is occupied by n_2 electrons, etc. This representation is motivated by the fact that occupation numbers is the only necessary information to construct a symmetric state. For an N -particle system, it is required that

$$\begin{aligned} \sum_i n_i &= N, \\ n_i &\geq 0. \end{aligned} \quad (15)$$

In the Hilbert space of an N -particle system, $|n_1, n_2, \dots\rangle$, subjecting to the above constraint, is a complete and orthonormal basis.

It turns out to be very useful to allow the system to change the number of particles. For this purpose, **Fock space** is defined by the *direct sum* of the Hilbert space of 0, 1, 2, ... particles. To hop between Hilbert space of different particles, we define the **creation operator**:

$$\hat{a}_i^\dagger |\dots, n_i, \dots\rangle = \sqrt{n_i + 1} |\dots, n_i + 1, \dots\rangle, \quad (16)$$

which connects the Hilbert space of an N -particle system to the Hilbert space of an $(N + 1)$ -particle system. The Hermitian conjugate of the creation operator is the **annihilation operator**:

$$\begin{aligned} \hat{a}_i |\dots, n_i, \dots\rangle &= \sum_{n'_i=0}^{\infty} |\dots, n'_i, \dots\rangle \langle \dots, n'_i, \dots | \hat{a}_i | \dots, n_i, \dots \rangle \\ &= \sum_{n'_i=0}^{\infty} |\dots, n'_i, \dots\rangle \langle \dots, n'_i + 1, \dots | \dots, n_i, \dots \rangle \sqrt{n_i + 1} \\ &= \sqrt{n_i} |\dots, n_i - 1, \dots\rangle, \end{aligned} \quad (17)$$

connecting the Hilbert space of an N -particle system to the Hilbert space of an $(N-1)$ -particle system. A special case is that $\hat{a}_i|\dots, 0, \dots\rangle = 0$.

The commutation relation between creation and annihilation operators are particularly simple

$$\begin{aligned} [\hat{a}_i, \hat{a}_j] &= 0, \\ [\hat{a}_i^\dagger, \hat{a}_j^\dagger] &= 0, \\ [\hat{a}_i, \hat{a}_j^\dagger] &= \delta_{ij}. \end{aligned} \quad (18)$$

With the above knowledge, the occupation number representation can be written as

$$|n_1, n_2, \dots\rangle = \frac{1}{\sqrt{n_1! n_2! \dots}} (\hat{a}_1^\dagger)^{n_1} (\hat{a}_2^\dagger)^{n_2} \dots |0\rangle, \quad (19)$$

where $|0\rangle$ is defined as the no-particle state $|0, 0, \dots\rangle$. Notice the difference between $|0\rangle$ and 0.

Operators in Second Quantization

The particle number operator is defined as

$$\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i, \quad (20)$$

it is easy to verify that

$$\hat{n}_i |\dots, n_i, \dots\rangle = n_i |\dots, n_i, \dots\rangle, \quad (21)$$

counting the number of particles in state i . The total number of particles is counted by

$$\hat{N} = \sum_i \hat{n}_i. \quad (22)$$

We now discuss general **one-body operators**. The basic principle to derive the form of operators in the second quantization formalism is that it should match the properties of the operator in the first quantized version. In occupation representation, the operators that we can define are generally physical observables summed over all particles:

$$\hat{T} = \sum_{i,j} t_{ij} \sum_{\alpha}^N |i\rangle_{\alpha} \langle j|_{\alpha}, \quad (23)$$

where i and j label states. An example is the total kinetic energy:

$$\frac{1}{2m} (\hat{\mathbf{p}}_1^2 + \hat{\mathbf{p}}_2^2 + \dots) = \sum_{\mathbf{p}} \frac{\mathbf{p}^2}{2m} \sum_{\alpha}^N |\mathbf{p}\rangle_{\alpha} \langle \mathbf{p}|_{\alpha}, \quad (24)$$

where the summation of i and j reduces to the summation of $\mathbf{p}$ as we have used the eigenstate of $\hat{\mathbf{p}}$. Generally,

$$t_{ij} = \langle i | \hat{T} | j \rangle, \quad (25)$$

where $\hat{t}$ is the operator corresponding to $\hat{T}$ with just one particle. To derive the second quantization form of the $\hat{T}$ operator, we act $\sum_{\alpha} |i\rangle_{\alpha} \langle j|_{\alpha}$ on the occupation number state (assuming $i \neq j$):

$$\begin{aligned}
\sum_{\alpha} |i\rangle_{\alpha} \langle j|_{\alpha} |\dots, n_i, \dots, n_j, \dots\rangle &= \sum_{\alpha} |i\rangle_{\alpha} \langle j|_{\alpha} \hat{S}_+ |i_1, i_2, \dots\rangle \frac{1}{\sqrt{n_1! n_2! \dots}} \\
&= \hat{S}_+ \sum_{\alpha} |i\rangle_{\alpha} \langle j|_{\alpha} |i_1, i_2, \dots\rangle \frac{1}{\sqrt{n_1! n_2! \dots}} \\
&= n_j \sqrt{n_i + 1} \frac{1}{\sqrt{n_j}} |\dots, n_i + 1, \dots, n_j - 1, \dots\rangle \\
&= \hat{a}_i^{\dagger} \hat{a}_j |\dots, n_i, \dots, n_j, \dots\rangle.
\end{aligned} \tag{26}$$

Similar equation can also be proved for $i = j$, yielding

$$\hat{T} = \sum_{i,j} t_{ij} \hat{a}_i^{\dagger} \hat{a}_j. \tag{27}$$

This is the general form of a one-body operator.

Alternatively, the above expression can be argued from a **change of basis**. For example, if there are two single-particle basis $\{|i\rangle\}$ and $\{|\lambda\rangle\}$, there transformation is

$$|\lambda\rangle = \sum_i |i\rangle \langle i|\lambda\rangle. \tag{28}$$

Therefore,

$$\begin{aligned}
\hat{b}_{\lambda}^{\dagger} &= \sum_i \hat{a}_i^{\dagger} \langle i|\lambda\rangle, \\
\hat{b}_{\lambda} &= \sum_i \hat{a}_i \langle \lambda|i\rangle.
\end{aligned} \tag{29}$$

To derive the expression for the operator $\hat{T}$, we note that if λ is the eigen-basis of $\hat{t}$,

$$\begin{aligned}
T &= \sum_{\lambda} t_{\lambda} \hat{b}_{\lambda}^{\dagger} \hat{b}_{\lambda} \\
&= \sum_{\lambda, i, j} \hat{a}_i^{\dagger} \langle i|\lambda\rangle t_{\lambda} \langle \lambda|j\rangle \hat{a}_j \\
&= \sum_{i, j} t_{ij} \hat{a}_i^{\dagger} \hat{a}_j.
\end{aligned} \tag{30}$$

We now discuss **two-particle operators**. The most common two-body operator take the form of

$$\hat{F} = \frac{1}{2} \sum_{\alpha \neq \beta} f(\hat{\mathbf{r}}_{\alpha}, \hat{\mathbf{r}}_{\beta}), \tag{31}$$

and can be written as

$$\hat{F} = \frac{1}{2} \sum_{\alpha \neq \beta} \sum_{i, j, k, m} \langle i, j | f | k, m \rangle |i\rangle_{\alpha} \langle j|_{\beta} \langle k|_{\alpha} \langle m|_{\beta}, \tag{32}$$

where

$$\langle i, j | f | k, m \rangle = \int d\mathbf{r} \int d\mathbf{r}' \varphi_i^*(\mathbf{r}) \varphi_j^*(\mathbf{r}') f(\mathbf{r}, \mathbf{r}') \varphi_k(\mathbf{r}) \varphi_m(\mathbf{r}'). \quad (33)$$

Since

$$\begin{aligned} \sum_{\alpha \neq \beta} |i\rangle_\alpha |j\rangle_\beta \langle k|_\alpha \langle m|_\beta &= \sum_{\alpha \neq \beta} |i\rangle_\alpha \langle k|_\alpha |j\rangle_\beta \langle m|_\beta \\ &= \sum_{\alpha, \beta} |i\rangle_\alpha \langle k|_\alpha |j\rangle_\beta \langle m|_\beta - \delta_{kj} \sum_{\alpha} |i\rangle_\alpha \langle m|_\alpha \\ &= \hat{a}_i^\dagger \hat{a}_k \hat{a}_j^\dagger \hat{a}_m - \delta_{kj} \hat{a}_i^\dagger \hat{a}_m \\ &= \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_k \hat{a}_m, \end{aligned} \quad (34)$$

therefore,

$$\hat{F} = \frac{1}{2} \sum_{i, j, k, m} \langle i, j | f | k, m \rangle \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_k \hat{a}_m. \quad (35)$$

Fermions

Fermions have antisymmetric state, and are therefore different from bosons by the extra sign factor. This factor is crucial because occupation numbers no longer uniquely specify the state.

It is convenient to write the fermionic many-body state as determinants known as **Slater determinant**:

$$\hat{S}_- |i_1, i_2, \dots, i_N\rangle = \frac{1}{\sqrt{N!}} \begin{vmatrix} |i_1\rangle_1 & |i_1\rangle_2 & \dots & |i_1\rangle_N \\ \vdots & \vdots & \ddots & \vdots \\ |i_N\rangle_1 & |i_N\rangle_2 & \dots & |i_N\rangle_N \end{vmatrix}. \quad (36)$$

By the properties of the determinant,

$$\hat{S}_- |i_1, i_2, \dots, i_N\rangle = -\hat{S}_- |i_2, i_1, \dots, i_N\rangle, \quad (37)$$

which is the manifestation of the Pauli exclusion principle.

We define the creation operator by

$$S_- |i_1, i_2, \dots, i_N\rangle = \hat{a}_{i_1}^\dagger \hat{a}_{i_2}^\dagger \dots \hat{a}_{i_N}^\dagger |0\rangle, \quad (38)$$

which immediately leads to

$$\{\hat{a}_i^\dagger, \hat{a}_j^\dagger\} = 0, \quad (39)$$

where $\{\}$ denotes anti-commutation. Anti-commutation immediately leads to $(\hat{a}_i^\dagger)^2 = 0$, another manifestation of the Pauli exclusion principle.

The occupation representation relies on a specific choice of order of single-particle states. Suppose such an order has been selected,

$$|n_1, n_2, \dots\rangle = (\hat{a}_1^\dagger)^{n_1} (\hat{a}_2^\dagger)^{n_2} |0\rangle, \quad (40)$$

where $n_i \in \{0, 1\}$. The effect of the creation operator is

$$\hat{a}_i^\dagger |n_1, n_2, \dots\rangle = (1 - n_i) (-1)^{\sum_{j<i} n_j} |\dots, n_i + 1, \dots\rangle. \quad (41)$$

and the effect of the annihilation operator is

$$\hat{a}_i |\dots, n_i, \dots\rangle = n_i (-1)^{\sum_{j<i} n_j} |\dots, n_i - 1, \dots\rangle. \quad (42)$$

The proof is similar to what has been done for bosons. It is easy to verify the canonical anti-commutation relation:

$$\begin{aligned} \{\hat{a}_i^\dagger, \hat{a}_j^\dagger\} &= 0, \\ \{\hat{a}_i, \hat{a}_j\} &= 0, \\ \{\hat{a}_i, \hat{a}_j^\dagger\} &= \delta_{ij}. \end{aligned} \quad (43)$$

It is easy to verify

$$\sum_{\alpha} |i\rangle_{\alpha} \langle j|_{\alpha} = \hat{a}_i^\dagger \hat{a}_j, \quad (44)$$

and the operators are now represented by

$$\begin{aligned} \hat{T} &= \sum_{i,j} t_{ij} \hat{a}_i^\dagger \hat{a}_j, \\ \hat{F} &= \frac{1}{2} \sum_{i,j,k,m} \langle i, j | f | k, m \rangle \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_m \hat{a}_k. \end{aligned} \quad (45)$$

Special attention needs to be paid to the order of the creation and annihilation operators.

Question 1. Can you ignore the sign and define

$$\hat{a}_i^\dagger |n_1, n_2, \dots\rangle = (1 - n_i) |\dots, n_i + 1, \dots\rangle? \quad (46)$$

Answer: yes, and sometimes useful. This is **Jordan-Wigner transformation**. Jordan-Wigner transformation is useful in 1D but usually not higher dimensions.

Applications in Tight-Binding

We focus on electrons (fermion). Without second quantization, the Hamiltonian is

$$\hat{H} = \sum_{\alpha} \frac{\hat{\mathbf{p}}_{\alpha}^2}{2m} + \sum_{\alpha} V_{\text{ext}}(\mathbf{r}_{\alpha}) + \sum_{\alpha \neq \beta} V(\mathbf{r}_{\alpha} - \mathbf{r}_{\beta}), \quad (47)$$

where the first term is the kinetic energy, the second term is external potential, and the third term is electron-electron interactions.

Tight-binding approximation assumes, due to strong external potential, the electrons mainly live within a Hilbert space of orbitals (mostly surrounding atoms). Therefore, the single-particle states are the orbitals. According to the rules of second quantization, the kinetic energy and the external potential can be written as $\sum_{i,j} t_{ij} \hat{a}_i^\dagger \hat{a}_j$, where

$$t_{ij} = \int d\mathbf{r} \varphi_i^*(\mathbf{r}) \left[\frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{r}) \right] \varphi_j(\mathbf{r}). \quad (48)$$

The name tight-binding comes from the approximation that the orbitals are very localized, such that $t_{ij} \neq 0$ only when the distance between orbital i and orbital j is small enough.

The electron-electron interaction part is $\sum_{i,i',j,j'} U_{ii'jj'} \hat{a}_{i\sigma}^\dagger \hat{a}_{i'\sigma'}^\dagger \hat{a}_{j\sigma} \hat{a}_{j'\sigma'}$, where

$$U_{ii'jj'} = \int d\mathbf{r} \int d\mathbf{r}' \varphi_i^*(\mathbf{r}) \varphi_{i'}^*(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \varphi_j(\mathbf{r}) \varphi_{j'}(\mathbf{r}'). \quad (49)$$

Here, we included the spin degree of freedom for each orbital, labeled by σ and σ' .

We now discuss some important interaction terms:

- **Direct density-density interaction:**

$$\sum_{i \neq j} V_{ij} \hat{n}_i \hat{n}_j, \quad (50)$$

where $V_{ij} = U_{ijij}$ and $\hat{n}_i = \sum_{\sigma} \hat{a}_{i\sigma}^\dagger \hat{a}_{i\sigma}$.

- **Exchange coupling:**

$$\sum_{i \neq j, \sigma, \sigma'} U_{ijji} \hat{a}_{i\sigma}^\dagger \hat{a}_{j\sigma'}^\dagger \hat{a}_{i\sigma'} \hat{a}_{j\sigma} = - \sum_{i \neq j} \tilde{J}_{ij} \left(\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \frac{1}{2} \hat{n}_i \hat{n}_j \right), \quad (51)$$

inducing ferromagnetic interaction between neighbouring orbitals.

- **Hubbard Interaction:**

$$\sum_{i, \sigma, \sigma'} U_{iiii} \hat{a}_{i\sigma}^\dagger \hat{a}_{i\sigma'}^\dagger \hat{a}_{i\sigma'} \hat{a}_{i\sigma} = \sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}. \quad (52)$$

Question 2. Which of the above interaction terms are large? Which are small?

Answer: Hubbard > Density-Density > Exchange.

The Hubbard Model

Including only the Hubbard interaction, the Hubbard model reads

$$\hat{H} = \sum_{i,j} t_{ij} \hat{a}_i^\dagger \hat{a}_j + \sum_i U \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}. \quad (53)$$

Hubbard model is the most important model in strong correlated condensed matter physics. At arbitrary filling, the model is unsolved. At half filling for large U , the model prefers anti-ferromagnetic order. At one less electron away from half filling for large U , Nagaoka proved that the system, with infinite interaction and certain conditions on the lattice, exhibit ferromagnetism.

The anti-ferromagnetic order at half filling is easy to understand. To keep it simple, we assume there are only two orbitals, such that there are only six states: $|\uparrow, \downarrow\rangle$, $|\downarrow, \uparrow\rangle$, $|\uparrow, \uparrow\rangle$, $|\downarrow, \downarrow\rangle$, $|2, 0\rangle$, $|0, 2\rangle$. We take the large U limit such that $|\uparrow, \downarrow\rangle$, $|\downarrow, \uparrow\rangle$, $|\uparrow, \uparrow\rangle$, $|\downarrow, \downarrow\rangle$ have the lowest energy. Their energy are the same with only the interaction terms. However, if kinetic term is included, $|\uparrow, \downarrow\rangle$, $|\downarrow, \uparrow\rangle$ can connect to $|2, 0\rangle$, $|0, 2\rangle$ to lower its energy, such that antiferromagnetism is favored.

Research Project

Using Hartree-Fock calculations to show that 120° -antiferromagnetic state is the mean-field ground state of Hubbard model on a triangular lattice for large enough U .